

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2162

Binary 138 kV
May 27, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2162. The Interconnection Customer proposes a 141.2 MW battery energy storage facility interconnecting at the Binary 138 kV substation in the Virginia area. The request was received on August 17, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 531 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$71,259,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Pahokee – St Clair 138kV	Greening 345/138 kV XFMR	154.7	225	28.4
Pahokee – Zephyrhills 138kV	Greening 345/138 kV Auto	137.1	272	28.4
Grant St – Daleville 138kV Line	New Denver Substation – Martin Dam 138kV	136.7	272	66.8
Binary – Grant Street 138kV	Martin Dam – New Denver 138kV	135.9	279	66.8
York Canyon – Binary 138kV	New Denver – Martin Dam 138kV	132.2	255	33.2
Aquasco – St Clair 138kV	Greening 345/138kV TX	120.4	245	10.7
Zephyrhills – Lamping Sta 138kV	Greening 345-138 kV Autotransformer	118.6	322	28.4
Aquasco – Crain Highway 138kV L	Greening 345/138kV Autotransformer	116.5	241	10.7
Crain Highway – Bauer 138kV	Greening Substation 345-138 kV Auto	106.2	231	10.7
Acec Lewiston – Indian Flat 138kV	Northern Neck – Chico A 138kV Line	104.0	209	7.9
Dunnsville – Northern Neck 138kV	Chico A – Northern Neck 138kV	100.1	215	5.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pahokee – St Clair 138kV	Rebuild line with higher-capacity conductor	\$21,529,000	44% (\$9,436,000)
Pahokee – Zephyrhills 138kV L	Rebuild line with higher-capacity conductor	\$6,906,000	44% (\$3,027,000)
Daleville – Grant Street 138kV	Rebuild line with higher-capacity conductor	\$8,958,000	67% (\$5,997,000)
Grant St – Binary 138kV	Rebuild line with higher-capacity conductor	\$3,634,000	67% (\$2,433,000)
York Canyon – Binary 138kV	Rebuild line with higher-capacity conductor	\$36,316,000	50% (\$18,231,000)
Aquasco – St Clair 138kV	Rebuild line with higher-capacity conductor	\$4,888,000	100% (\$4,888,000)
Zephyrhills – Lamping Sta 138kV	Rebuild line with higher-capacity conductor	\$9,127,000	44% (\$4,000,000)
Aquasco Substation – Crain Highway 138kV	Rebuild line with higher-capacity conductor	\$17,292,000	100% (\$17,292,000)
Crain Highway – Bauer 138kV	Reconductor line	\$541,000	100% (\$541,000)
Acec Lewiston – Indian Flat 138kV	Reconductor line	\$3,054,000	100% (\$3,054,000)
Dunnsville – Northern Neck 138kV	Reconductor line	\$2,360,000	100% (\$2,360,000)
	Total allocated to project		\$71,259,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.